



SIGNALLING AND PEER EFFECTS: THE MICROECONOMICS OF INCLUSION

MSc in HRM Dissertation

Supervisor: Dr. Nicola J. Bowman

Submitted to The Open University Business School in fulfilment of Master of Science in HRM
(CIPD Accredited)



ORCID QR code
<https://orcid.org/0009-0006-3363-9603>
Connecting research and researchers

Foreword and Acknowledgements

I am forever grateful to Almighty God, my strength, my fortress, my redeemer. My senior pastors, deacons and ministers, my parents, and friends for being so supportive of this endeavour I have been called to pursue. To my course mates, I thank you for endless support along the way through all the module TMAs and late nights working on submissions. Throughout the experience of postgraduate studies at *The Open University* (OU), I have been supported by my supervisor Nicola, several internal departments, external organisations and numerous people in academia who have worked endlessly to craft my critical understanding of Human Resource Management and who have helped refine my research topic around SHRM, inclusion and diversity, and wellbeing. These include *Library Research Support*, *Student Support* and of course my former OU advisers and current OU tutors. I would also like to take this opportunity to convey my gratitude and a tremendous thank you to all of those who have helped finance my graduate learning experience and research endeavours, either through tuition fee loans or through bursaries and grants, particularly I would like to focus my thanks on the efforts of the *British Academy of Management* (BAM), the Chair and Executive Committee of the *Royal Economic Society*, the *Chartered Institute for Personnel and Development* (CIPD) & its Trustees, the Student Fees team at *Student Finance England* (SFE) and the Finance Managers at the *Open University Student Budget Account* (OUSBA) who have helped me settle into distance learning and funded my pursuit throughout my time as a postgraduate student.

Declaration

I declare that this dissertation was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text. This work has not been presented for any other degree or professional certification.

Signed _____

Dated _____

DO NOT COPY

THESIS TITLE (*Final Draft Version*):

"Signalling and Peer Effects: The Microeconomics of Inclusion"

Alfred. A. B. Mayaki¹

Author Correspondence

Open University Business School
The Open University
Milton Keynes, MK7 6AA
Buckinghamshire, England
United Kingdom

Abstract

By hypothesising that SHRM signalling practices as organizational cues are designed to communicate SHRM initiatives to workers, this paper assesses N=1,034 survey responses using a two-staged least squared (2SLS) regression model and instrumental variable (IV) approach. This paper presents a study into the stability of an equilibrium between firm profitability given outlay on wages under optimality constraints and a SHRM signal using data sampled on expected wage demands given level of education.

Keywords: Information design, signalling, corporate communication, 2SLS, quantitative methods, econometrics, instrumental variable (IV)

1 | Introduction

Artificial Intelligence (Henceforth "AI") has been argued to carry the potential to change or affect various types of skill-based worker allocations, to mitigate the consequence of overexposure to unconscious biases, and to help design more comprehensive signalling approaches to allocate talent, and efficiently categorise new workers to roles, whilst developing an overarching organisational focus on *skill-based* diversity (Ramroop, 2025) more generally. However, according to Kolotilin, *et. al.* (2025), a peer effects problem exists in the research on AI's application to HRM, particularly with respect to the task of startup staffing processes (Abraham *et.al.*, 2025) and public sector recruitment (Chilunjika *et. al.*, 2022; Vogel *et. al.*, 2024). Gaps certainly do exist in the form of strategic choices available to supervised learning algorithms (Li *et. al.*, 2020) concerning the contextual bandit problem which may affect worker allocations and transfers (Harzing, 2001), however, one of the more pronounced gaps

in the literature on AI is competition and employment balances for neurodivergent and female workers (Iheonu *et. al.*, 2020, Mayaki, 2024; Mirowska and Arsenyan, 2025; Smith *et. al.*, 2025) which doesn't correct for the capture of those not in work or for sector-wide revenue effects.

Research Philosophy

The literature review will be in the form of the more legible narrative review. Unlike systematic reviews, which attempt to build upon scientific knowledge by analysing a wide variety of contributions to a specific field in a positivist way (The Open University, 2023), narrative reviews focus on the formation of compelling arguments using an *interpretivist* philosophy. However, later, in the Methodology as we shall come to see I will apply positivist techniques via the use of quantitative methods (a participant survey with an operative

¹All author correspondence regarding this thesis should be made to the following e-mail address: alfred.mayaki@ou.ac.uk. The author is a Royal Economic Society student member.

Likert scale component and a 2SLS regression with an instrumental variable).

Theoretical Background to the Topic

While AI and signalling are two very distinct topics (Lindgren, 2023), a great starting point is the work of Anton Kolotilin *et. al.* (2025) who defines two explicit cases of the ‘Kamenica-Gentzkow’ or ‘KG’ Bayesian persuasion model (Kamenica and Gentzkow, 2011; Che *et. al.*, 2022)—where many senders attempt to persuade a single receiver to take a particular action (to hire, given a revealed signal about their circumstances)—one case being the linear, and the other being the *nonlinear* (Yang, 2026) occurring in this respect, with a random walk² variable (Skardal and Adhikari, 2019). In a written work on signalling and its HRM context, another cohort of authors find evidence—in a review of digitalisation—that recruitment and selection is indeed mostly a *linear* process (Gentzkow and Kamenica, 2016) viewed in its *traditional* sense (Ilyushina, Walker and Timming, 2024). If we assume the linear case of Ilyushina *et. al.* (2024), i.e. a *frictionless* signalling environment or a case where the sender and receiver are able to successfully design a signal message that maximises the probability of signal realization, then the premise of this study is rather trivial, and the question is answered quite formally in the proofs in Kolotilin *et. al.*’s outstanding commentary. However, if we consider the nonlinear case *with costly frictions* (Che, Kim and Mierendorff, 2022) for both the sender and receiver, then the problem of peer effects becomes more complex.

Stakeholder Capitalism, Optimal Signalling and SHRM

Given nuances of the Stages model, otherwise known as the Business Life Cycle model, it is often the case that strategic narratives (Miles and Snow, 1984; Lengnick-Hall and Lengnick-Hall, 1988) of HRM tend to focus historically on unitarian forms of stakeholder capitalism especially under the resource-based view of HRM (Lado and Wilson, 1994), where the shareholder’s incentive and competitiveness are prioritised (Wright and Steinbach, 2022) but as Chapter 8 of Van Riel and Fombrun (2007) indicates, shareholders are not the only stakeholders in the organization

with incentive to act in the interests of the organization (what Lado and Wilson refer to as ‘proclivity’), workers too, can be viewed as *internal* stakeholders (Lesch *et. al.*, 2025). Signalling theory (Spence, 1973) therefore, widely seen as a seminal work (van Damme and Guth, 1991, p.264; Kolotilin, 2015), postulates itself as providing a microscopic lens into the origins of worker-firm matching in economic theory, demonstrating how *unobserved* productivity between workers and firms (Bafera and Kleinert, 2023) sets the scene for a cogent analysis of competing interests with respect to a many-to-one job market solution (Kolotilin *et. al.*, 2024; Ilyushina, Walker and Timming, 2024) where the peer effects problem arises. It remains the case that Spence (1973)’s main hypothesis—that signals act as communicative devices between firm and worker, such as when workers invest *more* in education signalling to firms that a higher wage is expected (Kolotilin, 2015)—is yet to be reconciled directly with the problem of peer effects and recent work on SHRM, and optimal worker allocation. The central challenge for this thesis is closing the gap between research which contextualises signalling theory in favour of the firm (Bafera and Kleinert, 2023) and in the context of the worker’s interests (Miller and Doyle, 2025).

Miller and Doyle on Reasonable Adjustments

How workers observe inclusive HRM strategies (Ramroop, 2025) both internally and externally in a VUCA environment (Barney, 1991) links to optimal signalling throughout the employee ‘*journey*’ (Gross, 2023) and workforce lifecycle (Celani and Singh, 2011; Miller and Doyle, 2025). The study investigates several reoccurring themes in the work of Celani and Singh (2011), Guest (2021) and Miller and Doyle (2025) with the premise of a research article on signalling theory forming a large part of the literature review. The study extends Miller and Doyle’s work by galvanising responses to a survey consisting of 11 questions³ and undertakes a 2SLS regression analysis (Greene, 2003; Kelejian, 1971; Chen, 2019), with an instrumental variable (IV).

² Notwithstanding the case of the *linear* random walk where the variable assumes independent and identically distributed properties

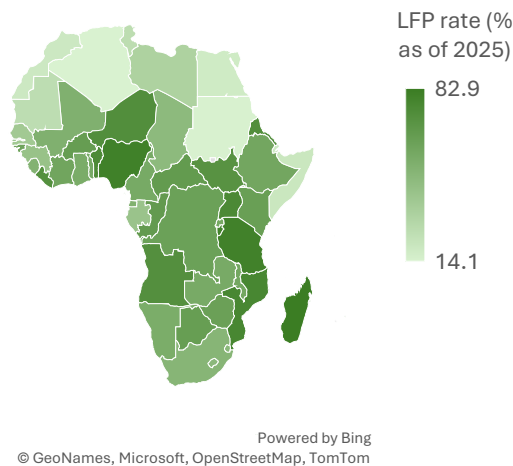
³ The participant survey’s core questions are available to view online: <https://alfredmayaki.me/msc-survey>

Female Labor Force Participation and Other Types of Inclusion

⁴In emerging economies such as West Africa, with the rise of AI, the central planner's problem arguably changes, from shifting the unemployment rate down to increasing the level of *female* labor force participation (Iheonu *et. al.*, 2020; Sarsons, 2024). Let us firstly assume a low female participation rate in an economy and that a hypothetical multinational firm F_N at time (t) , seeks to employ a cohort of n_M line managers, and n_N non-supervisory staff members at a new geographical location, with a given allocated personnel expense which we may refer to as a wage allowance or ex-ante wage budget $[B_N(t) \cdot (w)_N(t)]$. The shareholders of this

Fig 1: Female 'LFP' Indicator

Fig. 1: Female Labor Force Participation Rates* in Africa



organization $S_F(t)$ vote unanimously by motion that a cohort of line managers and non-supervisory workers should be closely managed by a team of expatriated and embedded human resource managers n_H in order to facilitate a geographical expansion of the firm into this new geographical location. New task roles and role profiles for all three categories of work (line managerial workers, non-supervisory staff, and human resource professionals) are identified, and a headcount figure is estimated for each role as part of a workforce plan.

Utility, Practices and Payoffs

The key model of this thesis defines a firm as consisting of a group of private shareholders $S_F(t)$, and a management team (including supervisory staff) denoted as $M(t)$, and a group of workers $N(t)$. Each worker in the firm has an average tenure of γ years in employment. As in Mayaki (2025) initial firm endowments are exogenously determined. The economy is assumed to be perfectly competitive. The firm's primary optimisation problem is initially a labor market allocation problem: that is, given an *estimated* annual turnover of $R_F(t)$, what should the optimal ex-ante wage budget $[B_N(t) \cdot (w)_N(t)]$ be set to in order to attract and ensure all three cohorts of workers are sufficiently skilled, productive, and that the firm earns a reasonable percentage of competitive market share in the first year of trading as a market entrant? We consider a case where there are working practices that dictate the value of workers to the supply chain. The solution to this labor market problem of allocation of female (and neurodivergent) workers is of course partly dependent on the size of the firm's wage budget $B_N(t) \cdot (w)_N(t)$, which is a function of the firm's estimated annual turnover $R_F(t)$. However, one might argue that strategic management, narrative⁵ and choice rather than simply the offered wage is the central consideration for incoming workers, company shareholders and senior management alike (Porter, 2008; Anderson and Markides, 2007).

Nonlinear Random Walks - Extending Kolotilin *et. al.* on Peer Effects

How then is *female* labor participation treated as a granular datapoint by firm $F_N(t)$? Prescriptively or normatively? Let us next assume the workforce is allocated in a *nonlinear* way to a firm $F_N(t)$ on the basis of productive output (Kolotilin *et. al.*, 2024) the cumulative distribution of which is strictly quasi-concave, non-decreasing, continuous and homogenous of degree 1 in w_N at time t (Lambotte, 2025: p. 620; Varian, 1992: p.72). We can assume also that there are a finite number of workers, each with preference utility $U_N(t)$ and an infinite number of allocations available to each firm in an economy $\sigma_F(t)$ and a finite number of priced transfers, which occur at a realised cost to each firm. Many-to-one

⁴ Source: OpenDataForAfrica.org

⁵ We use this term to generally imply the existence of any type of strategic planning within practice.

*This is a demographic measure of all eligible female adults (aged 15-64).

signalling occurs in regard to worker opportunity sets and working practices, which we denote as $P_N(t)$, which are either skill or competency-based.

Pre-Defining Equilibrium Conditions

Where contingency-based working practices foster collaboration, a high level of creative freedom under supervision and long-term appraisal and incentives reflecting groupwork, high-commitment working practices prioritise resources. Nevertheless, we only assume competency-based working practices display monotonicity along the firm *output* function, prioritising worker education, skill and productive efficiency. All three worker types may only signal to firms their level of experience and education which signal productive capacity and a categorical wage demand. We can therefore produce an objective function, where the real wage supplied to N workers $w_N^S(t)$ at time t is equal to the total firm revenue $R_F(t)$ minus the firm specific ex-ante wage budget divided by the total number of workers and multiplied by a firm specific *scalar* of working practices. Hence, equilibrium conditions are met where:

$$(1.11) \quad w_N^S(t) = w_N^D(t)$$

In Fig. 2 (below), the wage supplied by each firm is equal to the wage demanded by N workers which is slightly easier to deduce, as the function $w_N^D(t)$ is composed, as such, of

consumption-to-savings (Mayaki, 2024). The endogeneity in this paper's model relates to the worker's pre-employment signal about their level of education E_U and the worker's pre-employment signal about their skills capacity (level of experience) E_X multiplied by the likelihood that the worker is a member of a politically-affiliated trade union which is a percentage, $\mu_N(t)$, ranging between 0% (low likelihood of union affiliation) and 100% (certainty of union affiliation). Further empirical research may wish to explore the influence of peer effects on trade union membership under different equilibrium conditions.

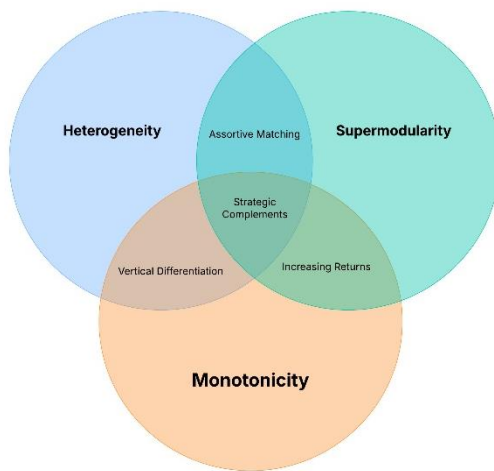
Further Equilibrium Conditions

In a repeated game of optimal information design, we are particularly interested in solutions where $w_N^S(t) \neq w_N^D(t)$ occurs or rather where heterogeneous worker allocations (which are ordered and ranked in group form based on education and experience) are not in equilibrium as per Blandin, Boyd and Prescott (2015) and as indicated by the outcome of the negotiated wage and productive output, where worker heterogeneity interjects with supermodularity and monotonicity of practices-to-transfers (Gentzkow and Kamenica, 2016; Kolotilin *et. al.*, 2025), hence the sweet spot occurs where the boundary condition of strategic complementarity (Galeotti *et. al.*, 2009: p.220) is always met (Fig. 2).

In Mayaki (2024), I argue for two things; the first being the existence of pre-game agreements by allowing for firms and workers to “discuss the *possibility* of entering into a contract for employment”, whereby players may also form incentivised network links with co-workers as with Lambotte (2025) prior to engaging in the signalling game (Blume and Ortmann, 2007: p.279), with the incentive of improving transfer capacity is measured by a shift in $w_N^D(t)$ where $w_N^D(t) > \sum_1^N w_N^S(t)$ occurs as the difference between signalled productivity and actual productivity. The second being a Walrasian market equilibrium *across* markets, where firms may ‘signal entry’ and ‘signal to workers’ about the nature of its respective competitive strategy and wage negotiation.

Fig 2: Three-Factor Venn Diagram

Fig. 2: Optimal Matching with Peer Effects



Kolotilin *et. al.* (2025)

exogenously determined concepts and variables such as preference for leisure and the ratio of household

2 | Relevant Literature Review

What exactly is an HRM strategy? What is so compellingly strategic about SHRM and how do workers

interact with such spuriously utilised lexicon to any real impact within the context of an organization? The answers are numerous. One insight comes from the work of David Guest has been prominent in HRM literature for well over 30+ years. In Guest (2021), the author studies the staff members of numerous bank branches in emerging economies and primarily defines signalling theory as ‘communication *by* and *within* an organisation’, in contrast to evidence from Laker *et al.* (2025). Citing Bowen and Ostroff (2004) as part of an integrated framework to study attribution theory, Guest (2021) concludes that signalling theory is to be viewed as a ‘*distinctive*’ approach to the HRM process. He acknowledges limitations to cross-sectional methodology and calls for longitudinal research approaches. The article by Bowen and Ostroff (2004) is one which touches upon multi-level relationships (Rousseau, 1985) between HRM and *organisational* performance through psychological “climate” as an example of higher-order social structures (other examples include organisational culture and organisational role structure, and organisational sensemaking (p.205). Bowen and Ostroff (2004) highlight the additional importance of ‘social context theory’ where formal roles are defined by organisational systems and ‘maintenance subsystems’.

Identifying Gaps in the Research Literature

The gap in the signalling literature is not very vast; however, it extends to recent work by Kolotilin *et al.* (2024) and with formal proofs with application to media censorship and later to diversely skilled workers and firms, requires further reading into recent work on AI and HRM to ascertain the proven hypotheses (Yang, 2026). There is little or no example, however, before Kolotilin *et al.* (2024) of any recent empirical results showing survey evidence achieved through *primary* research. Comparing Kolotilin *et al.* (2024) to van Damme and Guth (1989), the work of the latter and thereafter van Damme and Guth (1991) explicitly refers to Spence (1973) in a model focused on perfect equilibria—as previously demonstrated in Mayaki (2024) via ‘disagreement points’ in what van Damme and Guth refer to as ‘*uniformly* perfect equilibria’—worker and firm frictions are seen as a significant element to equilibrium search and matching models. Here, one might offset the pros of signalling by arguing that equilibrium analysis of corporate communication and information design in HRM is too hypercritical,

interested in detail which does not exist and is along the lines of the basic-hard approach, which fails to acknowledge the ‘H’ in HRM (Van Riel and Fombrun, 2007).

Critiquing Spence’s Signalling Game

In the view of Spence (1973) the notion of hiring is made analogous to the rationale of purchasing a ticket for a lottery draw, which itself is a problematic notion due to the euphemisms of risk, value and eventual loss being invoked. This rationale of adopting such a gamble is best described per a model composed by Barmby *et al.* (1994). Spence argues that there is often an element of informational asymmetry pertaining to full knowledge of the average workers’ skill or capability in the job prior to such a lottery concept being invoked. Assuming the lottery concept is associated with information that is rarely fully disseminated, a firm’s perception of this lottery gamble is what Spence (1973) credits as a “primary interest” due to the uncertainty involved in making the proposed hire when weighed against the wages offered by the firm to the worker as compensation for services rendered. When discussing the lottery, we speak here of a singular instance of a repeated game. The terms of reference adhered to by Spence, constitute very strict and formal ‘belief system’ for his term sheet mapping the firm’s perception of the lottery process. There are two kinds of observable attributes mentioned in Spence’s article: ‘indices’ and ‘signals’ (Spence, 1973; p. 357). The term ‘indices’ refers to immutable attributes observed by the firm amongst participants in the hiring ‘lottery’, attributes such as race or sex. This is followed closely by the familiar term ‘signals’, which references alterable attributes or those which can be manipulated by the worker, such as education and experience. For the purposes of Spence’s research, age is considered as an alterable attribute and is therefore seen to be an index. Signals and indices are seen to be ‘parameters’ in the conditional probability distribution reflecting the ‘beliefs’ of a firm. Upon reflection with some experience and exposure to high-level dialogue gained through my work with the Business in the Community (BITC) Opening Doors program (Impact in Business Recruitment, 2026), its fair and proper to infer proportionality. By this, I mean the Spencian idea that work is a lottery, is not an appropriate microscopic lens to view employment dynamics when compared to the objective of improving female education and labour participation (Iheonu *et al.*, 2020: p.261).

Synthesising Miller and Doyle and Spence's Signalling Game

Furthermore, much of the recommendations of the 2024 Autistica report authored in collaboration with Sir Robert Buckland KC on optimising a neurodivergent workforce can be supported by academic literature. If the Autistica report identifies the scenarios and problem faced by Autistic workers, then Miller and Doyle (2025) most certainly shed further light on this problematisation of a Spencian-styled 'lottery' scenario by providing what one might call a dialectical synthesis to the concerns highlighted in the report. According to Miller and Doyle (2025) the article mentions the theoretical framework of Khan *et. al.* (2023) whose literature review and empirical study on the topic is the only post-COVID era work used as authority, but the article also acknowledges that research in the area of inclusive signalling to neurodiverse workers is relatively limited. When extending Khan *et. al.* (2023)'s ideas for further research in a HR context, Miller and Doyle (2025) apply a systems-based analysis of 'signalling' theory (or rather a systematic approach based on academic literature) in the hiring and selection process to uncover a consummate view that the inclusion of the neurodiverse job worker in the workplace is paramount. Both co-authors argue organisations should go *beyond* the business case for diversity in recruitment by making the process itself clearer (Ramroop, 2025).

Critical Evaluation of Celani and Singh on Outcomes

One can divide the approach for achieving such an eventuality as follows. Firstly, the work of Miller and Doyle treats an interesting argument in favour of neuroinclusive approaches to hiring (Khan *et. al.*, 2023) by encouraging proactivity from the firm's managers. The line of thinking goes: If a firm provides a list of 'frequently requested adjustments' then this action is therefore a proactive signal to workers who may require such adjustments to participate in the hiring process. Likewise, in agreement with Ramroop (2025), by sufficiently educating recruiters and hiring managers involved in the act of recruitment and selection on what "neurodivergent *behaviours*" are, and by thereafter implementing a follow-up action: prioritising diversity, equity and inclusion over the Spencian view of hiring as a lottery, we arrive at a mechanism which appreciates neurodiversity as a recruitment priority. Critiques may argue this may create a

consequentialist dependence on prior training investment. In contrast to Miller and Doyle (2025)'s unrelenting pragmatism on the subject of signalling best practice, Celani and Singh (2011) have conducted an analysis of signalling theory from a *multi-level* perspective (with the use of 'contextual effects'). In multi-level critique, the assessment of how recruitment-related signals is seen to reveal unknown characteristics of the organization becomes *more* stratified (Guest, 2021). The theory posited provides links between individual and organisation-level signals from recruitment activities. Links are made between market signals and instrumental or symbolic worker inferences. Celani and Singh (2011) are concerned with outcomes (as opposed to the interest of Davies *et. al.* (2023) for example who are concerned with recruitment *methods* faced by Autistic adults. They therefore take up a consequentialist position (a position which derives itself from the moral philosophy of consequentialism) – with a focus on outcomes (Guest, 2021). By introducing their topic in such a way, Celani and Singh (2011) conclude by challenging readers via invitation to study a gap in the signalling literature—signalling from a recruiter perspective i.e. from the organisation's perspective. This means to imply that the organisation (recruiter or hiring manager) does indeed perceive and internalise what Spence (1973) defines as 'alterable' attributes (for instance, the highest level of education signalled by the worker or the worker's length of experience/number of promotions or career highlights). Signalling theory when viewed through the Spencian microscopic lens is therefore arguably either angst or incongruent (Smith *et. al.*, 2025) on the one hand, or catalyst and predicator (Lup and Canonico, 2024) on the other. By contrasting the approaches reviewed in Celani and Singh (2011) with that of Miller and Doyle (2025) it is possible to observe how in hindsight both authors take completely opposing perspectives on signalling in the context of inclusion. Chang and Chin (2018) offer a third way – idiosyncrasy. They define signalling as the communication of unique information, or rather "the sending out of information that competitors cannot easily imitate" analysing HRM signalling on firm websites over a three-year period. Signalling therefore becomes important in job markets where numerous firms compete to "*increase market share*" as Chang and Chin infer (2018) stressing the importance of firm actions and separating equilibrium.

The workforce lifecycle, otherwise known as the employee ‘*journey*’ (Gross, 2023) under competence-based HRM (Lado and Wilson, 1994) covers a vast amount of terrain including interviews (Mirowska and Arsenyan, 2025), which are unique to every firm yet even so, an important contemporary aspect with respect to signalling theory (Barba *et al.*, 2025) are the areas of performance management and compensation, which have increasingly come under the remit of *Large Language Models* (LLMs)—where the problem of algorithmic bias, which is almost always inherent to generative AI, continues to surface (Galdin and Silbert, 2025; p. 36). Because competency-based HRM is about optimising the utilisation of skill, signalling theory becomes paramount to the objective of allocating resources in the organization. In passing, Spence himself has condemned the “flooding of the system” with models that are trained poorly, in what Spence refers to categorically as models which are “rubbish” (Spence *et al.*, 2024: p. 1985) in a recent interview. As with social desirability bias (Licon, 2025), erroneous human performance reviews can be and are replicated by LLMs as centrality or even as leniency biases (Rilke and Sliwka, 2026: p. 16).

3 | Data & Methodology

Introduction

Making use a 10-point Likert scale (where 10 = strongly agree, 1 = strongly disagree, and 5 = neutral), this thesis surveys a total of 1,034 former (in-work) job market candidates across geographies using data gathered through multiple years of global content activity performed by the author. A 10-point Likert scale is justified because it is a quantitative and ordinal measure of respondent behaviours, attitudes and opinions. As stated earlier in this thesis, the purpose of the participant survey is primarily to ascertain how each respondent perceives the practice of inclusive interview accommodation/adjustments and specifically measures the meaning of this to the respondent by using a value scale from 0 to 10. Here, we place emphasis on the journey of the respondent as an interview candidate. Then, by using the *IMRAD* scientific research format (Shiely *et al.*, 2024), the methodology of the thesis applies guidance from the *Econometric Analysis 5th Edition* textbook (Greene, 2003: p.403) and prior mathematical inference published in a working paper by the author (Mayaki, 2024) to structure a

2SLS regression estimating the *nonlinear* value of the expected wage regressed upon education with endogeneity. We mimic the model from the linear case in Mayaki (2024: p.8) which outlines a ‘base case’ Pareto-Nash result with a Taylor approximation. Firstly, we regress a vector of the ‘*expected wage*’ variable E_t , occurring across all $N = 1,034$ participants in survey form (a response to a current salary question) at time t for all respondents given their ‘level of education’ (this is also a preliminary survey question based on undergraduate, postgraduate and PhD as optional responses), which is a variable proxy representing the survey response concerning one’s wage demands relative to one’s current level of education at time t , and secondly, an above market wage which firms have set to attract productive workers.

This discrepancy in wage expectations aligns accordingly with the utility of the worker discipline device model by Shapiro-Stiglitz, a theory employed in Mayaki (2024) describing how firm F ’s wage offer strategy (for the linear case i.e. homogenous workers) minimises the likelihood of excessive shirking (Shapiro and Stiglitz, 1984) or to a lesser extent, avoidance of pooled workers who may carry a probability of signalling low productivity levels (Barmby *et al.*, 1994). Therein lies the core premise of Kamenica-Gentzkow (KG) Bayesian persuasion framework (Che *et al.*, 2022). Table 3.1 (below) demonstrates these variables as one-half of a simultaneous equation system, where the problem is *always* to balance the incentives of the worker with the motivation of the firm in a long-run equilibrium, particularly with the given likelihood of a technology shock. In Table 3.1—which illustrates the second half of the simultaneous equation system—the model assumes that the firm observes (or has *full* disclosure of) the revealed productivity of current workers, the profitability of their business unit, and thereafter sets its wage in subsequent t periods.

Table 3.1: 2SLS Simultaneous Equations

3.1 – What is Being Estimated in the Study?

Key Variable	Variable
Expected wage given education	E_t
Firm observed wage-level data	W_t

Source: Imbens (2014: p.328)

The study then builds on the recently published work of Miller and Doyle (2025). The nature of the newly formatted participant survey questions (See Appendices) remains consistent with Miller and Doyle (2025)'s original survey after a transformation from open format to closed. The interrogation format was ultimately changed from open to closed to better quantify responses of workers. The ten new questions ascend from 1-10 in a Likert scale where 1 equates to a strong disagreement and 10 equates to the highest possible level of agreement across all questions. The forte of this format is it allows structural estimates to be derived when these survey variables are embedded into an objective function (a constrained maximisation problem). The two-stage least squared (2SLS) regression model (Greene, 2003; Kelejian, 1971; Chen, 2019) in Equation 3.1 is defined as the dependent variable: W_t representing the demanded wage of the worker, an intercept β_0 , an observed intertemporal level of education variable and its estimator β_1 and X_1 , and finally a residual or error term, ε .

Instrumental Variables Using 2SLS

In Table 3.1 the 2SLS model is defined as follows:

$$(3.1) \quad \text{Expected Wage } (E_t) = \text{Intercept } (\beta_0) + \beta_{1t} \text{Education}(X_{1t}) + \beta_{2t} \text{Dummy } (X_{2t}) + \varepsilon$$

In the 2SLS (Abraham, *et.al.*, 2025), the first stage (the instrument) identifies a dummy variable (Z) that is correlated with the HRM signal (the 'treatment') but is uncorrelated with the dependent variable's error term. For example, $Z = 1$ when a firm provides a list of frequently requested adjustments and $Z = 0$ when this list is not provided in the recruitment process. In the 2SLS second stage (the outcome), the study applies the predicted values from the first stage to estimate the impact on the dependent variable Y . In terms of variables, it was assumed that one key variable emerged as a primary source in the regression function, which is: the measure of worker education. The dependent variable Y is derived from a secondary source of data. These inferences, in part, form questions which in turn are separate to the research question and are therefore secondary in nature.

Table 3.2: The Methodology

Table 3.2: Further Methodological Approaches

Handling outliers and skewness	Post-normalisation <i>Competitiveness index</i>
Addressing assumption violations	If fails homoskedasticity Then robust estimation
Measurement scale adjustments	Ordinal (rank based) Interval (numeric)

The importance of the research survey should not be understated, because it seeks to enquire into how diverse potential workers perceive their roles in the recruitment process *prior to* being onboarded as employees (in other words, it asks how *neurodiverse* workers interact with established structures) through the proxy of website recruiting activities. The seminal applied instance of this research topic is the online ratings and HRM signalling paper pioneered by Chang and Chin (2018) who test evidence to investigate the hypothesis that three things are likely to occur prior to onboarding. Firstly, a high-commitment HRM strategy is pursued and that firms conduct recruitment online via a website. This assumes a finite degree of discretion is made on the part of the firm/sender. Secondly, the firm may initiate a signal to many receivers via a website. This signal communicates a list of reasonable adjustments (interview accommodations) which indicates the firm's high-commitment HRM strategy. This signal initiates an effect on the employees' online ratings.

Outlining the Research Hypothesis

The authors (Chang and Chin, 2018) argue for five important hypotheses which form the starting point for this study:

- **Hypothesis 1:** Firms with more high-commitment HRM will signal more HRM information on websites.
- **Hypothesis 2:** Firms signalling more HRM on websites will see a trend of more applications.
- **Hypothesis 3:** Firms' signalling of HRM fully mediates the effect of commitment HRM on recruitment.
- **Hypothesis 4:** Firms' *actual* HRM influences employees post-experience outcomes.
- **Hypothesis 5:** Firms' signalling of HRM will not increase an employees' online ratings if the effect of commitment HRM (on recruitment) is controlled.

Regarding the research hypothesis, this thesis is partly covered by *signalling* (Spence, 1973), *worker*

attraction (Dutta and Mishra, 2021), *strategic HRM* (Miles and Snow, 1984) and *labour market analysis* (Acemoglu, *et. al.*, 2008; Pissarides, 2017). My research will aim to hypothesise on a gap on the signalling topic by focusing on peer effects, capabilities, and resources, asking a question not currently addressed by the literature on signalling – how these factors affect competitive advantage given the specific context of SHRM. The following hypotheses are subject to supervisory changes and will be honed and amended as and when new information in the literature is discovered, reviewed and internalised. In order, these initial hypotheses are:

- **Hypothesis 1:** The average worker views the firm's intent to accommodate as inclusive.
- **Hypothesis 2:** Firms signalling high-commitment HRM practices have happier workers.
- **Hypothesis 3:** Firms signalling high-commitment HRM strategies tend to be more inclusive.
- **Hypothesis 4:** Firms harnessing the influence of AI can achieve high-commitment *and* be optimally inclusive.

The null hypothesis of no correlation for this paper implies there needs to be two questions asked. The first question is in a separating equilibrium with known payoffs, and standard assumptions (complete information and perfect monitoring), how do firms successfully navigate a many-to-one 'Spencian lottery' nonlinear equilibrium scenario, using the offered wage as an optimal signal or as a navigating mechanism? Secondly, but no less important, one must ask what the sub-game Pareto-optimal signal is, in a game of separating or pooling equilibrium (Kolotilin, 2024) a firm should send to all workers in order to be inclusive? Finally, at what level of signal does a firm who signals acceptance of a neurodiverse condition, become accepted by a diverse worker?

Justification for Selecting Topic

With respect to the selected subtopics, inclusion, signalling and peer effects; sufficient due diligence, care, and careful consideration were each observed when narrowing down a broader topic and selecting signalling as a topic of choice. There is also a need to resolve intertextual reference between the matching function description as found in Mayaki (2024) and its peripheral proposition on Nash

bargaining, invoked in retrospect of signalling theory, albeit in a nuanced and slightly muted manner. Not only is Spence's signalling game clearly an important canonical work to most if not all the relevant authors in the scope of the 2nd Chapter of this thesis but the signalling game serves to answer a question that is very topical in current affairs and geopolitical news – how will the world of work be recalibrated when the negative impact of AI on white-collar work increases unemployment and digital working practices by the organisation?

Specifying the Hypothesis Test

As is common for hypothesis tests that estimate two variables using this approach, we set the confidence interval at 95% with a null of no correlation, which will reject the null for all results below the p-value of 0.05.

REFERENCES

- Abraham, D., Galeotti, A., and Szeidl, A. (2025) 'Peer Effects in Entrepreneurship', *LBS Working Paper*
- Acemoglu, D., Kremer, M., and Mian, A. (2008). 'Incentives in markets, firms, and governments', *The Journal of Law, Economics, & Organization*, 24(2), 273-306.
- Anderson, J., and Markides, C. (2007). Strategic innovation at the base of the pyramid. *MIT Sloan management review*.
- Bafera, J., and Kleinert, S. (2023). Signaling Theory in Entrepreneurship Research: A Systematic Review and Research Agenda. *Entrepreneurship Theory and Practice*, 47(6), 2419-2464.
- Barba, G., Corallo, A., Lazoi, M., and Lezzi, M. (2025) "Large language models for competence-based HRM: A case study in the aerospace industry," *Journal of innovation & knowledge*, 10(5).
- Barmby, T., Sessions, J., and Treble, J. (1994). 'Absenteeism, Efficiency Wages and Shirking', *The Scandinavian Journal of Economics*, 96(4), pp. 561-566.
- Barney, J. (1991). 'Firm resources and sustained competitive advantage', *Journal of Management*, 17(1), 99.
- Blandin, A., Boyd, J.H., and Prescott, E.C. (2015). 'Equilibrium with Mutual Organizations in Adverse Selection Economies', *Federal Reserve Bank of Minneapolis, Research Department Working Paper* 717
- Blume, A., and Ortmann, A. (2007). 'The effects of costless pre-play communication: Experimental evidence from games with Pareto-ranked equilibria'. *Journal of Economic Theory*, 132(1), 274-290.
- Bowen, D. E., and Ostroff, C. (2004). Understanding HRM-firm performance linkages: The role of the "strength"

- of the HRM system. *The Academy of Management Review*, 29(2), 203–221.
- Celani, A., and Singh, P. (2011), "Signaling theory and applicant attraction outcomes". *Personnel Review*, 40(2), pp. 222–238
- Chang, E., and Chin, H. (2018). Signaling or experiencing: Commitment HRM effects on recruitment and employees' online ratings. *Journal of Business Research*, 84, 175–185.
- Che, Y.K., Kim, K., and Mierendorff, K. (2022) 'Keeping the Listener Engaged: a Dynamic Model of Bayesian Persuasion', *arXiv e-print*
- Chen, W. (2019) 'Income and Effort: An Instrumental Variables Approach' *Atlantic Economic Journal*, 47, pp. 485–497
- Chilunjika, A., Intauno, K. and Chilunjika, S.R. (2022) "Artificial intelligence and public sector human resource management in South Africa: Opportunities, challenges and prospects," *SA Journal of Human Resource Management*, 20(1), pp. e1–e12.
- Davies, J. *et al.* (2023) 'Access to employment: A comparison of autistic, neurodivergent and neurotypical adults' experiences of hiring processes in the United Kingdom', *Autism: the international journal of research and practice*, 27(6), pp. 1746–1763.
- Dutta, D., and Mishra, S.K. (2021) Predictors of applicant attraction among Gen-X and millennials: evidence from an emerging economy. *International Journal of Manpower*, 42(8), pp.1479–1499.
- Eppler, M., Hinne, A., and Bunzli, F. (2026). 'You Strategy Needs a Visual Metaphor', *Harvard Business Review*
- Galdin, A., and Silbert, J. (2025). 'Making Talk Cheap: Generative AI and Labor Market Signaling'. *arXiv preprint arXiv:2511.08785*
- Galeotti, A., Goyal, S., Jackson, M. O., Vega-Redondo, F., and Yariv, L. (2009). 'Network Games', *Review of Economic Studies*, 77(1), 218–244.
- Gentzkow, M. and Kamenica, E. (2011). Bayesian Persuasion. *The American Economic Review*, 101(6), 2590–2615.
- Gentzkow, M. and Kamenica, E. (2016) "A Rothschild-Stiglitz Approach to Bayesian Persuasion," *The American economic review*, 106(5), pp. 597–601.
- Greene, W.H. (2003) *Econometric analysis*. 5th ed. International ed. Upper Saddle River, N.J: Prentice Hall.
- Gross, N. (2023) 'Perceived Opportunities and Barriers on the Employee Journey', *UW-Stout Masters Thesis Collection*,
- Guest, D.E. *et al.* (2021) 'Signalling theory as a framework for analysing human resource management processes and integrating human resource attribution theories: A conceptual analysis and empirical exploration', *Human resource management journal*, 31(3), pp. 796–818
- Iheonu, C.O., Nwodo, O., Anaduaka, U., and Ekpo, U. (2020) "Inequality and female labour force participation in West Africa," *European journal of government and economics.*, 9(3), pp. 252–264.
- Ilyushina, N., Walker, M., and Timming, A.R. (2024) 'Digitalisation and the Future of HRM' Chapter in: Adisa, T.A. (ed.) "*HRM 5.0: Unpacking the Digitalisation of Human Resource Management*". 1st ed. 2024. Cham: Springer Nature Switzerland.
- Imbens, G. W. (2014). Instrumental Variables: An Econometrician's Perspective. *Statistical Science*, 29(3), 323–358.
- Jones, D.A., Willness, C.R. and Madey, S. (2014) 'Why Are Job Seekers Attracted by Corporate Social Performance? Experimental and Field Tests of Three Signal-Based Mechanisms', *Academy of Management Journal*, 57(2), pp. 383–404.
- Kelejian, H. H. (1971). Two-Stage Least Squares and Econometric Systems Linear in Parameters but Nonlinear in the Endogenous Variables. *Journal of the American Statistical Association*, 66(334), 373–374.
- Khan, M. H., Grabarski, M. K., Ali, M., and Buckmaster, S. (2023). Insights into Creating and Managing an Inclusive Neurodiverse Workplace for Positive Outcomes: A Multistage Theoretical Framework. *Group & Organization Management*, 48(5), 1339–1386.
- Kolotilin, A. (2015). Experimental design to persuade. *Games and Economic Behavior*, 90, 215–226.
- Kolotilin, A. and Li, H., and Zapechelnjuk, A. (2024). 'On Monotone Persuasion', *Discussion Papers 2024-05, School of Economics, The University of New South Wales*.
- Kolotilin, A., Corrao, R., and Wolitzky, A. (2025). 'Persuasion and Matching: Optimal Productive Transport', *Journal of Political Economy*, 115(4), 1334–1381
- Lado, A. A., and Wilson, M. C. (1994). 'Human resource systems and sustained competitive advantage: A competency-based perspective'. *Academy of Management Review*, 19(4), 699.
- Laker, B. *et al.* (2025) 'To Change Company Culture, Focus on Systems—Not Communication', *Harvard Business Review*,
- Lambotte, M. (2025), Peer Effects in Binary Outcomes: Strategic Complementarity and Taste for Conformity With Endogenous Networks. *J Appl Econ*, 40: 608–626.
- Lengnick-Hall, C.A. and Lengnick-Hall, M.L. (1988). 'Strategic human resources management: a review of the literature and a proposed typology', *Academy of Management Review* 13(3), 454–470.
- Lesch, L., Lachance, E. L., Kerwin, S., & Wicker, P. (2025). From gender policies and practices to organisational performance of sport governing bodies. *European Sport Management Quarterly*, 25(4), 538–558.

- Li, D., Raymond, L. R., and Bergman, P. (2020) 'Hiring as Exploration', *NBER Working Paper Series No. 27736*, perspectives. *Research in Organizational Behavior*, 7, 1–37
- Licon, J.A. (2025) "ChatGPT is Bullshit (Partly) Because People are Bullshitters," *Philosophy & technology*, 38(2).
- Lindgren, S. and Edward Elgar Publishing, publisher (2023) *Handbook of Critical Studies of Artificial Intelligence*. First edition. Edited by S. Lindgren. Northampton: Edward Elgar Publishing.
- Lup, D. and Canonico, E. (2024) 'Bridges and gatekeepers: Employees' willingness to refer qualified candidates on the autism spectrum', *Human Resource Management*, 63(6), pp.1025-1043
- Mayaki, A.A.B. (2024) 'Pareto-Nash Reversion Strategies: Three Period Dynamic Co-operative Signalling with Sticky Efficiency Wages', *SSRN Working Paper* (Self-reference)
- Mayaki, A.A.B. (2025) 'Pareto-Nash Allocations under Incomplete Information: A Model of Stable Optima' *SSRN Working Paper* (Self-reference)
- Miles, R.E. and Snow, C.C. (1984). 'Designing strategic human resources systems', *Organizational Dynamics* 13(1), 36–52.
- Miller, H. and Doyle, N. (2025) 'Signalling a diversity climate: Neurodivergent experiences and perceptions during selection and assessment', *Journal of occupational and organizational psychology*, 98(3).
- Morgan, H. (1996) 'Adults with Autism: A guide to theory and practice', Cambridge: Cambridge University Press.
- Mirowska, A. and Arsenyan, J. (2025), A Bridge Too Far: Signalling Effects of Artificial Intelligence Evaluation of Job Interviews. *International Journal of Selection and Assessment*, 33: e70008.
- Park, C.C. (1998) "Adults with Autism: A Guide to Theory and Practice," *Psychiatric services (Washington, D.C.)*. American Psychiatric Publishing, pp. 1105–1105.
- Pissarides, C. A. (2017). *Equilibrium Unemployment Theory*. 2nd edn. Cambridge, MA: MIT Press.
- Porter, M. E. (2008). The five competitive forces that shape strategy. *Harvard business review*, 86(1), 78.
- Ramroop, M. (2025) *Building inclusion: a practical guide to equity, diversity and inclusion in architecture and the built environment*. Abingdon, Oxon: Routledge.
- Rilke, R. and Sliwka, D. (2026) 'When Algorithms Rate Performance: Do Large Language Models Replicate Human Evaluation Biases?', *IZA Discussion Paper 18371*,
- Roehl, M. T. (2019). The impact of SHRM on the psychological contract of employees: A typology and research agenda. *Personnel review*, 48(6), 1580-1595.
- Rousseau, D. M. (1985). Issues of level in organizational research: Multi-level and cross-level perspectives. *Research in Organizational Behavior*, 7, 1–37
- Sarsons, H. (2024), How the other half works: Claudia Goldin's contributions to our understanding of women's labour market outcomes. *Scand. J. of Economics*, 126: pp. 419-439.
- Shapiro, C., and Stiglitz, J. E. (1984). Equilibrium Unemployment as a Worker Discipline Device. *The American Economic Review*, 74(3), 433–444.
- Shiely, F., Gallagher, K., and Millar, S. R. (2024). How, and why, science and health researchers read scientific (IMRAD) papers. *PLOS ONE*, 19(1), e0297034.
- Skardal, P. S., and Adhikari, S. (2019). Dynamics of nonlinear random walks on complex networks. *Journal of Nonlinear Science*, 29(4), 1419-1444.
- Smith, A., van Wagoner, H. P., Keplinger, K., and Celebi, C. (2025) 'Navigating AI Convergence in Human–Artificial Intelligence Teams: A Signaling Theory Approach', *Journal of Organizational Behaviour*, 0:1-31
- Spence, M. (1973) "Job Market Signaling," *The Quarterly journal of economics*, 87(3), pp. 355–374.
- Spence, M., Behar, A. and Jayadev, A. (2024) "Artificial Intelligence in the Age of Uncertainty: Michael Spence in Conversation with Anurag Behar and Introduced by Arjun Jayadev," *Information systems frontiers*, 26(6), pp. 1971–1988
- The Open University (2023) 'Concerning Sustainable HRM, Ethics and Stakeholder Theory', Personal Blogs: *The Labour Economics Blog*
- Van Damme, E., and Guth, W. (1989) 'Equilibrium selection in the Spence signalling game', *CentER Discussion Paper*, Unknown Publisher,
- Van Damme, E., and Güth, W. (1991). Equilibrium Selection in the Spence Signaling Game. In: Selten, R. (eds) *Game Equilibrium Models II*. Springer, Berlin, Heidelberg.
- Van Riel, C.B.M., and Fombrun, C.J. (2007). *Essentials of Corporate Communication: Implementing Practices for Effective Reputation Management* (1st ed.). Routledge.
- Varian, H. R. (1992) 'Microeconomic Analysis: Third Edition', London & New York: Norton and Company, Inc.
- Vinayak, P. C., Khan, B. M., and Jain, M. C. (2017). Role of signalling theory in potential applicant attraction: a firm branding perspective, *International Journal of Emerging Research in Management and Technology*, 6(4), pp. 230-233
- Vogel, D., Döring, M. and Sievert, M. (2024) 'Motivational signals in public sector job advertisements and how they relate to attracting and hiring candidates, *Public Management Review*, 26(10), pp.2868-2900.
- Wright, P.M. and Steinbach, A.L. (2022) 'Pivoting after almost 50 years of SHRM research: toward a

stakeholder view,' *Asia Pacific journal of human resources*, 60(1), pp. 22–40.

Yang, M.M. (2026). "Mitigating Disability Bias in Hiring: The Role of Inclusion-Focused Generative AI in Complex HR Decisions," *Human Resource Management Journal*: 1–13.